Project Report

Investigation of Gene Expression Regulation Using PhET Interactive Simulation

Student Name: P. Prathipa Sriji

Date: June 8, 2026

Platform: PhET Interactive Simulations, University of Colorado

Simulation: Gene Expression Essentials

Project Type: Independent Virtual Laboratory Investigation

1. Abstract

This project investigates the molecular mechanisms regulating gene expression using the PhET Interactive Simulations platform. Through systematic manipulation of transcription factors, RNA polymerase affinity, and mRNA degradation rates across three simulation modules, the study demonstrates how protein synthesis is controlled at the transcriptional level. Real-time quantitative data from the Multiple Cells simulation revealed natural variability in protein levels across cell populations, consistent with biological stochasticity.

2. Introduction

Gene expression is the fundamental process by which genetic information encoded in DNA is converted into functional proteins. This process involves two key stages: transcription (DNA to mRNA) and translation (mRNA to protein). Regulation of gene expression is critical for cellular function, development, and response to environmental stimuli.

The objective of this project was to investigate how molecular regulators--specifically transcription factors and RNA polymerase binding affinity--control the rate of protein synthesis using an interactive virtual simulation environment.

3. Materials and Methods

3.1 Platform

Software: PhET Interactive Simulations (HTML5)
Simulation: Gene Expression Essentials
Source: University of Colorado, Boulder
URL: https://phet.colorado.edu

3.2 Experimental Modules

Expression	- Visualize transcription and translation at single-gene level
mRNA	- Monitor mRNA production dynamics over time
Multiple Cells	- Analyze protein level variability across cell populations

3.3 Variables Tested

Positive transcription factor concentration (None to High)
Negative transcription factor presence (Absent/Present)
RNA polymerase affinity for DNA (Low to High)
mRNA degradation rate

3.4 Procedure

1. Loaded Gene Expression Essentials simulation
2. Tested Gene 1, Gene 2, and Gene 3 individually
3. Adjusted Positive Transcription Factor concentration and observed RNA polymerase binding
4. Introduced Negative Transcription Factor and recorded expression blockage
5. Modified RNA polymerase affinity slider and measured transcription rate changes
6. Switched to Multiple Cells module
7. Set cell population to 'Many' and ran 30-second simulation
8. Recorded Average Protein Level vs. Time graph data
9. Adjusted Concentration, Affinities, and Degradation parameters and repeated

4. Results

4.1 Expression Module Observations

Condition	RNA Polymerase Binding	Protein Output
Baseline (no factors)	Moderate	Low
High Positive TF	Strong, sustained	Increased
Negative TF present	Blocked or reduced	Decreased/None
High RNA Pol Affinity	Rapid initiation	Faster accumulation

4.2 mRNA Module Observations

mRNA levels correlated directly with transcription factor presence
Higher positive factor concentration produced steeper mRNA accumulation curves
mRNA degradation rate inversely affected steady-state mRNA levels

4.3 Multiple Cells Module - Quantitative Data

Graph Generated: Average Protein Level vs. Time (0-30 seconds)

Parameter Setting	Protein Level Trend
Default	Gradual increase with fluctuations
Increased Concentration	Steeper upward trend
Increased Affinities	Faster initial rise
Increased Degradation	Lower steady-state level

Key Observation: Protein levels across multiple cells showed inherent variability (noise) around the mean, reflecting stochastic gene expression in biological systems.

5. Discussion

The results confirm that gene expression is regulated at multiple levels:

1. Transcriptional control: Positive transcription factors enhance RNA polymerase recruitment to promoter regions, increasing transcription initiation frequency. Negative transcription factors competitively inhibit this binding, demonstrating repression mechanisms.

2. Binding affinity: RNA polymerase affinity for DNA determines how efficiently transcription begins. Higher affinity mimics strong promoter sequences in natural systems.

3. Cellular variability: The Multiple Cells simulation revealed that even genetically identical cells produce proteins at different rates due to random molecular interactions. This stochasticity is a well-documented phenomenon in molecular biology and has implications for drug response variability and single-cell analysis.

These findings align with established molecular biology principles and demonstrate the utility of virtual simulations for exploring complex biological systems without physical laboratory access.

6. Conclusion

This project successfully investigated gene expression regulation using interactive simulation technology. Key conclusions:

- Transcription factors are primary regulators of protein synthesis rates
- RNA polymerase affinity directly impacts transcription efficiency
- Protein expression varies naturally across cell populations
- Virtual simulations provide effective platforms for hypothesis-driven molecular biology investigation

7. Skills Developed

Skill Category	Specific Competency
Experimental Design	Variable control, hypothesis testing
Data Analysis	Real-time graph interpretation, quantitative observation
Molecular Biology	Transcription, translation, gene regulation
Scientific Reporting	Structured documentation, evidence-based conclusions
Digital Literacy	Simulation-based scientific inquiry

8. References

PhET Interactive Simulations. (2026). Gene Expression Essentials. University of Colorado, Boulder.

https://phet.colorado.edu
Alberts, B. et al. (2015). Molecular Biology of the Cell (6th ed.). Garland Science.

9. Appendix: Screenshots

Screenshots documenting experimental stages are retained as evidence of project completion.

Submitted by: P. Prathipa Sriji

Date: June 8, 2026